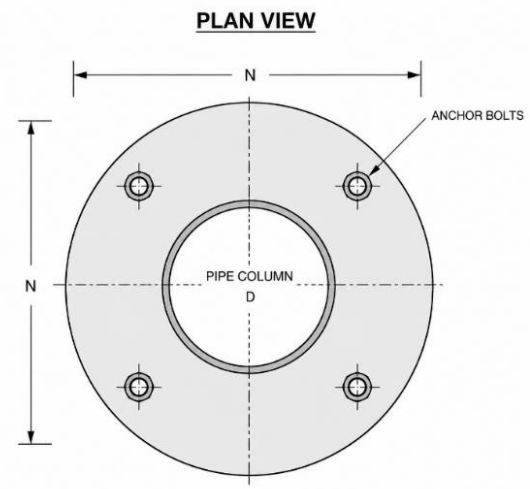
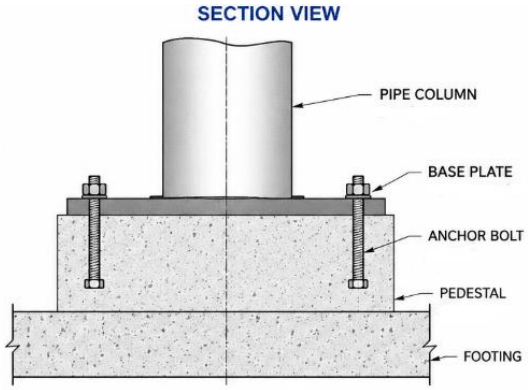
2. Base plate segment

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	1637.5732 kips	Main Tank Design (Predefined)
Governing overturning moment (LRFD)	M_u	5673.437 kip-ft	Main Tank Design (Predefined)
Factored Shear Force (LRFD)	V_u	57.22 kips	Main Tank Design (Predefined)
Pipe diameter	D_{pip}	20.04 in	Main Tank Design (Predefined)
Number of anchor bolts	N_b	4	From Anchor Bolt Design
Anchor bolt diameter	d_b	1.5 in	From Anchor Bolt Design
Anchor bolt hole diameter	d_b	2 in	From Anchor Bolt Design
Bolt arrangement dimensions	b×l	12 in × 12 in	From Anchor Bolt Design
Base-Plate cross-section	D_{co}	30	
Pedestal cross-section	P×L	39 in × 39 in	From Anchor Bolt Design
Concrete compressive strength	f'_c	4 ksi	User Input
Steel yield strength	F_y	36 ksi	User Input
Modulus of elasticity	E	29,000 ksi	AISC Material Property
LRFD resistance factor (plate bending)	ϕ_b	0.90	AISC LRFD
Required base plate thickness	t_{bp}	To be determined	AISC Base Plate Design



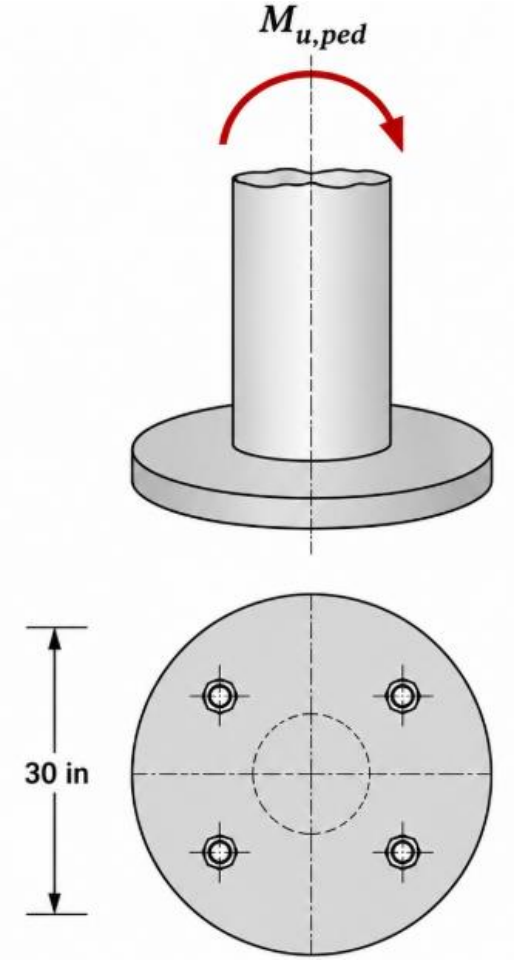


1. Determine the Factored Moment.

$$\mathrm{M_{u,ped}} = \frac{M_u}{\mathrm{N_{ped}}}$$

$$\mathrm{M_{u,ped}} = \frac{9455.728}{4}$$

$$\mathrm{M_{u,ped}} = 2{,}363.93 \text{ kip-ft}$$



Convert to kip-in:

$$\mathrm{M_{u,ped}} = 2{,}363.932(12)$$

$$\mathbf{M_{u,ped} = 28{,}367.184 \text{ kip-in}}$$

2. Pedestal Area

$$\mathrm{A_2} = (39)(39)$$

$$\mathrm{A_2} = 1521 \text{ in}^2$$

Base Plate Area

$$\mathrm{A_1} = \frac{\pi D^2}{4} = \mathrm{A_1} = \frac{\pi(30)^2}{4}$$

$$\mathrm{A_1} = 706.86\ in^2$$

Compute Bearing Stress

$$\mathrm{F_p} = 0.85\phi_c f'_c \sqrt{\frac{\mathrm{A_2}}{\mathrm{A_1}}}$$

$$F_p = 0.85(0.65)(4)\sqrt{\frac{1521}{706.86}}$$

$$F_p = 3.24 \text{ ksi}$$

Check Limit

$$1.7\phi_c f'_c = 1.7(0.65)(4) = 4.42 \text{ ksi}$$

$$3.24 < 4.42 \ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

3. Assume Base Plate Cross-Section

Adopt:

$$N = B = 30 \text{ in}$$

Base plate area:

$$A_1 = \frac{\pi(30)^2}{4}$$

$$A_1 = 706.86 \text{ in}^2$$

Check Bearing Capacity

$$P_n = F_p A_1$$

$$P_n = (2.87)(706.86)$$

$$P_n = 2029 \text{ kips}$$

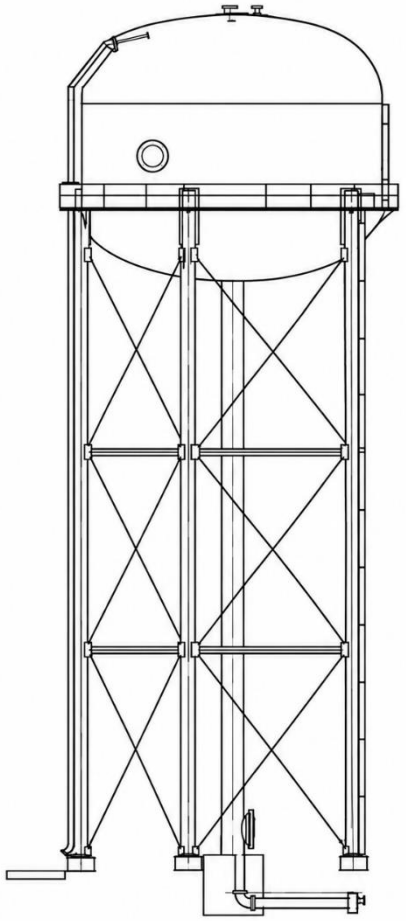
Applied load:

$$P_u = 929.741 \text{ kips}$$

Note: for this scenario we considered half of the pedestals in compression and half of them in tension as show in the photo

$$2029 > 929.741$$

PASS



4. Determine Equivalent Eccentricity

$$e = \frac{M}{P}$$

Substitute:

$$e = \frac{17020.31}{464.871}$$

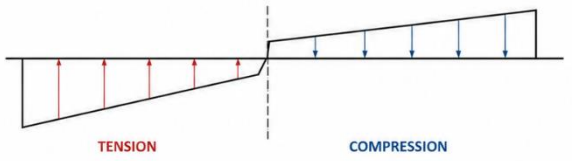
$$e = 36.613 \text{ in}$$

Check Bearing Condition

For full compression:

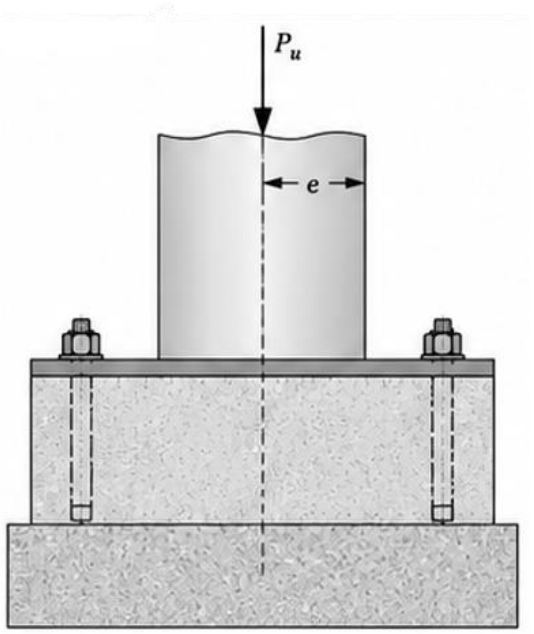
$$e \leq \frac{N}{6}$$

$\frac{N}{6} = \frac{30}{6} = 5.0$ in……. if this cause controls Pu/(1.0*N)



Since:

$36.613 > 5.0$……..if this cause controls (Pu/0.5*N)



Full compression does not occur.

Therefore:

- Uplift develops.
- Anchor bolts resist tension.
- Partial compression exists beneath the base plate.

$$\boxed{e = 36.613 \text{ in} > \frac{30}{6} = 5.0 \text{ in}}$$

5. Determine Resultant Anchor Bolt Force

From the completed anchor bolt design:

$$T_{leg} = 172.774 \text{ kips}$$

Number of anchor bolts:

$$N_b = 4$$

Anchor force per bolt:

$$T_{bolt} = \frac{T_{leg}}{N_b} = \frac{172.774}{4} = 43.193\ kips$$

6. Determine Base Plate Thickness

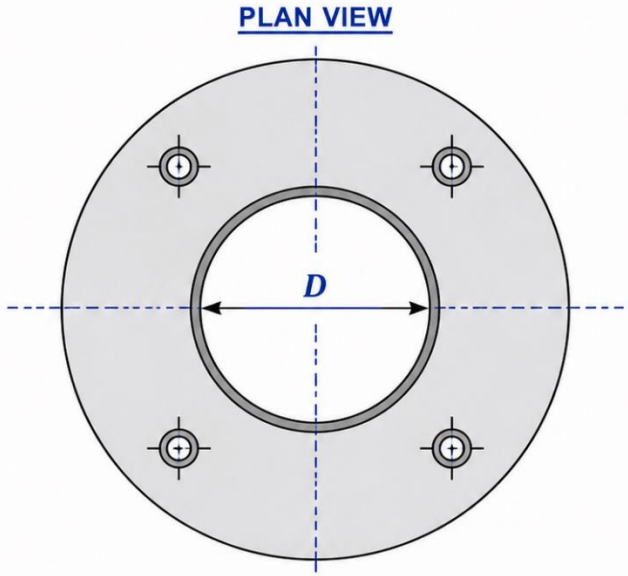
6.1. Determine Bearing Area of Pipe Column

Cylinder diameter:

$$D = 1.67 \text{ ft}$$

$$D = 1.67(12)$$

$$D = 20.04 \text{ in}$$



Column bearing area:

$$A_b = \frac{\pi D^2}{4}$$

$$A_b = \frac{\pi(20.04)^2}{4}$$

$$A_b = 315.42 \text{ in}^2$$

6.2. Determine Bearing Pressure

$$q = \frac{P_u}{A_b}$$

$$q = \frac{464.871}{315.42}$$

$$q = 1.473 \text{ ksi}$$

$$\mathbf{q = 1.473 \text{ ksi}}$$

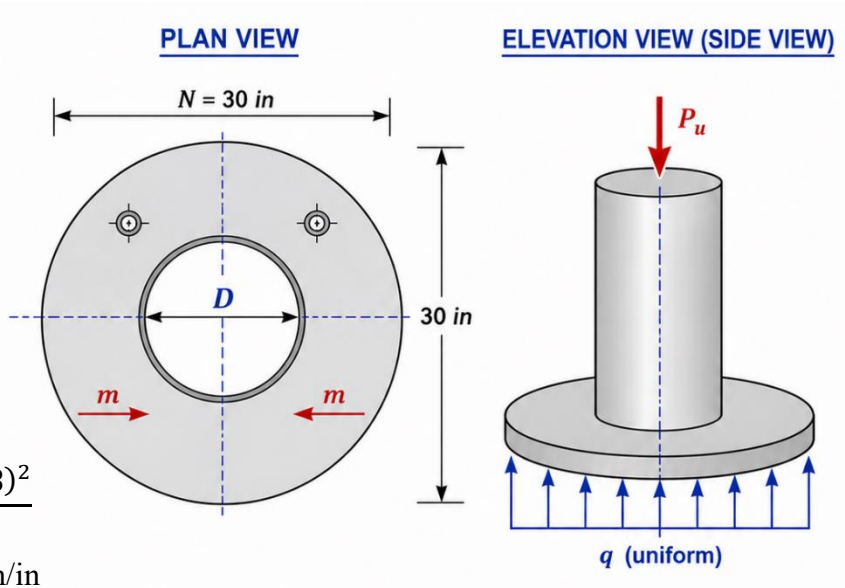
6.3. Determine Plate Projection

$$m = \frac{N - D}{2}$$

$$m = \frac{30 - 20.04}{2}$$

$$m = 4.98 \text{ in}$$

$$\mathbf{m = 4.98 \text{ in}}$$



6.4. Determine Plastic Moment per 1-in Strip

$$M_{plu} = \frac{qm^2}{2}$$

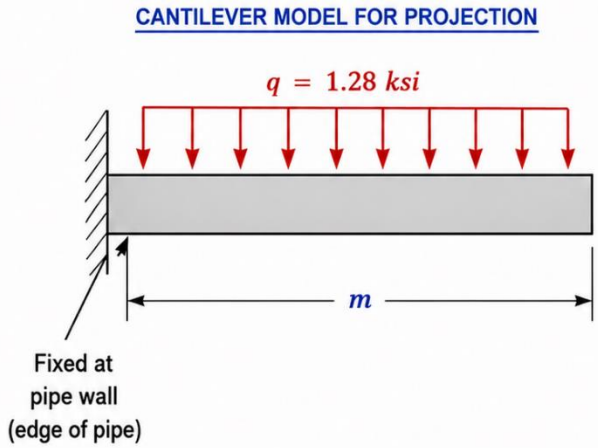
$$M_{plu} = \frac{(1.28)(4.98)^2}{2}$$

$$M_{plu} = 15.87 \text{ kip-in/in}$$

$$M_{plu} = 15.87 \text{ kip-in/in}$$

6.5. Determine Required Thickness

Using AISC LRFD:



$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4(15.87)}{0.90(36)}}$$

$$t_p = \sqrt{\frac{63.48}{32.40}}$$

$$t_p = \sqrt{1.959}$$

$t_p = 1.40$ in $\mathbf{t_p = 1.40 \text{ in}}$ Adopt: $\underline{\mathbf{t_p = 1.50 \text{ in}}}$